ENN595-1 Project Students
Semester Two 2026

Investigation via the Engineering Literature
Assessment 1A

Submit: Electronically, via Canvas Assessment Upload (Week 5)

Student Name: ____

Student No.: ____

Research Topic Title: ____

Project Supervisor: ____

Assessment Tasks

1. Draw and label a concept map of your research topic;
2. Translate those key concepts into a search statement;
3. Search for relevant engineering literature in a library database;
4. Identify authoritative, reputable and reliable information sources relevant to your research topic;
5. Submit a summary analysis and evaluation of three (3) key references;
6. Locate three (3) published authorities (acknowledged experts) in your field of research;
7. Identity, categorize and synthesise trends and directions found in the literature;
8. Explain what impact the literature findings have had on the design of your project;
9. Compile a reference list of the most useful literature on your topic.

This assessment item will be marked by your project supervisor.

On completion of this assessment, you should have formed an information research strategy that will allow you to discover evidence based scholarly, trade and professional literature that will underpin your research project proposal.

Guidance is available from the QUT **Library website**, including relevant library **Researching by study area** guides.

Advice is available from the HiQ service point (V block level 3), Online Help (Chat or email) or consultation (by appointment) with your engineering librarians.

1. Concept Map : Draw and label a graphical outline view of your research question.

Firstly, describe the overall project objective and list any questions to be investigated. Include any scope considerations, for example, location or time-based.

Concept maps are graphical tools that organize knowledge and assist you to clarify the concepts and show the relationship between concepts. For further information click here.

Use the space below to develop your concept map.

Now take the terms and phrases from your concept map to develop and refine a search strategy.

2. Develop a search statement.

Separately identify the **key concepts** you wish to search for. Click here and here for further information.

1st Concept		2nd Concept		3rd Concept
	AND		AND	

Now list any synonyms (words with the same or similar meaning) that may also retrieve relevant information. List here any spelling variations e.g., colour / color ; tire / tyre ; harbour / harbor.

	Synonyms		Synonyms		Synonyms	
OR		AND		AND		OR

As an alternative to the Boolean **OR** search operator, you can (where appropriate) opt to truncate a search keyword. Apply the keyword truncation symbol (usually an asterisk *) to the stem of a word to simultaneously retrieve the plural form, plus synonymous variant right-hand grammatical endings of your search keyword.

e.g., spectr will retrieve spectra, spectral, spectrometry, spectroscopy, spectrum, spectrums.*
List those of your search keywords that can be usefully truncated in the box below:

Note that if the database supports autostemming, as Compendex, Scopus and Web of Science do by default, truncation is not required. If you do truncate the terms, that is also okay.

However, with Web of Science, autostemming is not performed when a phrase is enclosed in quotation marks. If you want to stem the word or words in a phrase search, you will need to use the truncation symbol – typically *.

Structure your search statement in the box below using **Boolean** search logic operators **(AND, OR, NOT).** Use parentheses ( ) so that you can order a **keyword** search. Enclose a literal text string in inverted commas to force an exact key **phrase search**. For example:
("artificial intelligence" OR "fuzzy logic") AND (power AND (distribution OR transmission))

3. Search for relevant engineering literature in a library database.

Name the most useful literature database searched: Identify this QUT Library subscribed database (and if applicable, its delivery platform). For example: QUT Library Search; Compendex (on Engineering Village); Web of Science; Scopus.

Database Name: ______________________________

Database Platform: ______________________________

Comment on the scope of your chosen database and why it proved to be of high value to your literature search:

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4. Identify relevant and authoritative information sources.

List key **information sources** applicable and relevant to your research topic. Attempt to include at least one in each of the following three categories:

a. Peer-reviewed journal titles / Conference publications – the eng. capstone guide will assist.

b. Texts / Handbooks / Encyclopaedias – the eng. capstone guide will assist.

c. Standards (or regulations) / Patents – the finding Standards & Patents guides will assist.

5. Evaluate and analyse three (3) key references.

From your search results (typically these will be journal articles or conference papers or book chapters) you may use the template below to evaluate and analyse key references.

Reference # 1 -- Use **IEEE Citation Style** (or an alternative style agreed with your supervisor)
Citation:
How does this paper relate to my topic?
What key findings are reported?
How does this paper relate to other work in the field?
Interpret important figures/tables/illustrations (cite the page number).

List other references to investigate further (a good paper can often lead to another).
General notes/observations on this paper's attributes.

Reference # 2 – Use **IEEE Citation Style** (or an alternative style agreed with your supervisor)
Citation:
How does this paper relate to my topic?
What key findings are reported?
How does this paper relate to other work in the field?
Interpret important figures / tables / illustrations (cite the page number).

List other references to investigate further (a good paper can often lead to another).
General notes/observations on this paper's attributes.

Reference # 3 -- Use **IEEE Citation Style** (or an alternative style agreed with your supervisor)
Citation:
How does this paper relate to my topic?
What key findings are reported?
How does this paper relate to other work in the field?
Interpret important figures / tables / illustrations (cite the page number).
List other references to investigate further (a good paper can often lead to another).

General notes/observations on this paper’s attributes.

6. Identify and locate published authorities (experts) on your research question.

Name up to three (3) published authors (external to QUT) who are **clearly experts on your topic**. State why you believe they are an expert. You can usually judge this from the extent and quality of research material they have published and/or the extent to which their research papers have been cited. If it is given, list their latest institutional affiliation (i.e., where they work). **Provide a hyperlink** to your expert's academic or professional web page or their ORCiD ID that lists their recent publications, their experience and engagement with the research question/problem. Should they hold a relevant, registered patent, then state that.

7. Identify, categorize and synthesize findings from your examination of the literature.

Categorise major findings (issues, trends or themes) apparent in your preliminary investigation of the literature.

8. Explain the impact of your literature evaluation on the design of your research project.

Describe and explain what impact these findings might have on the direction, focus or scope of your research project.

9. **Reference List.**

Cite using IEEE Citation Style (or an alternative style agreed with your supervisor) at least eight (8) of the most relevant and useful references you have read on your project topic.